American Board of Family Medicine

2023 IN–TRAINING EXAMINATION

SCORE RESULTS HANDBOOK

2023 In-Training Examination Score Results Handbook

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American Board of Family Medicine, Inc. (ABFM®)

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Dear Program Director:

Thank you very much for participating in this year's In-Training Examination (ITE). As you know, the ITE provides the opportunity for you to gauge the academic progress of the residents in your program. It also provides residents with an opportunity to simulate the experience of taking the certification examination and given the psychometric comparability to the Family Medicine Certification, get an estimate of the likelihood of passing the exam via the Bayesian Score Predictor on RTM.

Please note that you will be able to access your program's results via the RTM. This handbook is intended to provide you with statistical analyses and explanations that will help you assess how the residents in your program performed and allow comparison to other programs. If you need assistance regarding your results or how to access them, please do not hesitate to contact our Support Center at 877-223-7437 or Kathy Botner at 859-429-7360. We appreciate that you have always utilized this information in the way that it has been intended to be used. Please do not hesitate to contact us if you should have any questions about the material in this handbook.

We look forward to working closely with you in our continuing efforts to improve all of our certification activities. At the Residency Leadership Summit (formerly PDW), we will have a workshop on using our tools to help guide development of residents. We have also created a toolkit of resources to help you and your residents in understanding and navigating ABFM Family Medicine Certification—and look forward to getting your feedback at about how that can be more useful. We look forward to talking with you.

Thank you for all you do.

Sincerely yours,

Warren Newton, M.D., MPH
President and Chief Executive Officer

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Introduction

This handbook provides information on how to interpret the reports associated with this year's *In-Training Examination* (ITE). These reports provide performance information about individual residents, the residency programs, and the nation. We would like to thank the Association of Family Medicine Residency Directors (AFMRD) and all the program directors whose substantive input guided the development of these reports. There are six sections to this document.

1) Resident-level reports and their interpretation.
2) Program-level reports and their interpretation.
3) Data files for both individual and program level data.
4) Instructions for manual scoring.
5) Bayesian Score Predictor use and interpretation.
6) Psychometric summary

SECTION I focuses on the interpretation of the individual resident's reports. Examples of the individual resident's *Examination Performance Report* and *Item Performance Reports* are provided, along with an explanation of each area of the report.

SECTION II provides examples of program-level reports. The *Mean Scaled Score* bar graph shows the program's average score by training year alongside the national average for that training year. The *8-Year Trend Chart* also provides program-to-national score comparisons for each program year of residency and the overall performance for the last eight years.

SECTION III provides resident-level subtest information (data files). Please note that these subtests are on the same scale (200-800) as the overall scaled score.

SECTION IV provides instructions for scoring the examination manually. It includes a list of any questions that were deleted from scoring, a raw score to scaled score conversion table, and a roster of which content category each question represented.

SECTION V explains what the Bayesian Score Predictor is and how to use it. The Bayesian Score Predictor can be found on the ABFM's website (https://rtm.theabfm.org/bayesian/predictor).

SECTION VI provides some summary information regarding the psychometric performance of the ITE on this administration. This could be useful to researchers.

If you have any questions, please contact us:

General questions	Support Center	**(877) 223-7437**	help@theabfm.org
Using the RTM System	Kathy Botner	**(859) 429-7360**	kbotner@theabfm.org
Scoring / Bayesian Score Predictor	Ting Wang Ph.D.	**(859) 554-2093**	twang@theabfm.org
Scoring / Bayesian Score Predictor	Thomas ONeill Ph.D.	**(859) 687-2538**	toneill@theabfm.org

In-Training Exam Summary Statistics

Table 1 provides summary statistics for the 2023 ITE. The statistics include the number of Accreditation Council on Graduate Medical Education (ACGME) accredited residency programs participating in the ITE, as well as the average size of the residency program and the associated standard deviation. Results from international residency programs are not included in this document.

Table 1. Summary Statistics for the 2023 ITE

Total Number of Programs Participating:	719
Mean Participating Residency Size:	20.5
Standard Deviation of Residency Size:	9.8

Table 2 provides a breakdown of resident's reported scaled scores by year of residency. As expected, the scaled score values tend to increase as residents advance through their program. The averages for PGY1 to PGY3 ranged from 347 to 427, a difference of 80 scaled score points. Program year 1 to program year 2 exhibits the greatest increase in scores, as indicated by an improvement of 50 scaled score points. Please note that PGY4 includes only residents who are in their 4th year of residency and are participating in the Length of Training (LoT) pilot study. PGY4 does not refer to residents who are in a dual specialty program (family medicine & psychiatry, family medicine & internal medicine, etc.)

Table 2. Residents' Reported Scaled Scores on the ITE 2023 by Year of Residency.

Year of Residency	Total Test	N	Mean	Standard Deviation
1	Scaled Score	5,158	347.0	73.6
	Percent Correct	5,158	58.2	6.2
	Raw Score	5,158	115.2	12.2
2	Scaled Score	4,851	396.8	74.6
	Percent Correct	4,851	62.2	6.0
	Raw Score	4,851	123.2	11.9
3	Scaled Score	4,666	427.4	76.3
	Percent Correct	4,666	64.7	5.9
	Raw Score	4,666	128.1	11.7
4	Scaled Score	35	482.3	70.9
	Percent Correct	35	68.9	5.2
	Raw Score	35	136.5	10.2
Total	Scaled Score	14,710	389.2	82.0
	Percent Correct	14,710	61.6	6.6
	Raw Score	14,710	122.0	13.1

SECTION I

Resident-level Reports

Overview

For each resident that takes the ITE, the residency program will receive two resident-level reports related to the resident's individual performance. Residents can download their own reports by accessing their physician portfolio on the ABFM website. The first report, the *2023 In-Training Examination Performance Report* (Figures 1a & 1b), presents overall performance and performance within specific content areas. This report closely resembles the format of the score reports used in the Family Medicine Certification Examination (FMCE). This is intended to familiarize residents with the type of feedback they will receive following their board certification examination. The first page of this three-page report explains how to interpret the results, while the next two pages represent the resident's performance broken out by the blueprint categories (body systems) and by clinical categories.

The second resident-level report, the *Resident's 2023 Item Performance Report* (Figures 2a and 2b), groups the questions by organ system (blueprint category) and by clinical content area and then indicates whether the question was answered correctly or incorrectly. Those questions answered incorrectly by the resident are shown with bold red font, whereas those answered correctly are shown with regular green font.

In-Training Examination Score Report

Header. Please refer to the sample report in Figures 1a & 1b. The score report provides resident-level performance information on the 2023 ITE. At the top of the report, you will see (1) the resident's ABFM ID# along with (2) their name. You will also see the (3) Mean Scaled Score for Year 1 [or Year 2 or Year 3 as appropriate], (4) Scaled Score, and the (5) Number of Unanswered Items.

Mean Scaled Score for PGY_. This represents the mean scaled score for those residents who are in the same year of residency (1, 2, or 3) as the resident who is receiving the report. Comparing an individual resident's score to the appropriate mean can provide a gross indicator of how the candidate performed relative to other residents with comparable amounts of training.

Scaled Score. This represents an estimate of how the resident would perform on the FMCE if they had taken it instead of the ITE. Presently, the minimum passing standard on the certification examination is 380. If the resident's score on the ITE was just barely 380, they would be expected, theoretically, to have only a 50-50 chance of passing the certification examination. A score of 440 would increase the probability of passing to over 90%.

Of course, it is important to consider that there are going to be other intervening factors that can affect the outcome on the certification examination as well. For example, a resident's score on the certification examination could increase due to additional study and training or the score could decrease if the resident reduced the amount of study time in the months before the examination. Similarly, stressful events (marriage, divorce, serious illness, death of a loved one, etc.) could impede a resident's ability to study in a focused manner and subsequently have a negative impact.

Although the intent is to connect the results from the ITE to the FMCE, it is important to keep in mind that the ITE is not a replacement for the FMCE. The certification examination enjoys a much higher level of security in its administration, has more rigorous item review processes, and has a higher degree of precision due to the larger number of questions administered and the targeting of those questions. To place this in appropriate perspective, appreciate that the fees charged for each examination are reflective of ABFM's relative costs for developing each, approximately $65 for the ITE versus $1300 for the FMCE.

Score Report Body

The lower portion of each score report has four columns and a bar chart. Both the *Organ System/Blueprint Category* chart (Figure 1a) and the *Clinical Category* chart (Figure 1b) have the same columns and similar bar charts. The only difference is that a different content category structure is used to classify the questions.

Content Area. The first chart uses the Organ System/Blueprint Category structure to organize the results. The second chart organizes the results by Clinical Category. Although a question may be theoretically related to more than one content area, each question is categorized as being in only one content area within a chart.

Exam Blueprint Percentage. The exam blueprint percentage is the percentage of questions that are specified by the FMCE's test plan specifications. This information might be useful for deciding what content areas to focus on when preparing for the certification exam. The second chart (Figure 1b), where the results are presented by clinical category, does not specify the percentage of questions that will be included on the exam by clinical category because the examination is not assembled by clinical category, only by major organ system. Also, be aware that that the percentage of questions in each category on a particular version of the ITE might be off by a percentage point or two.

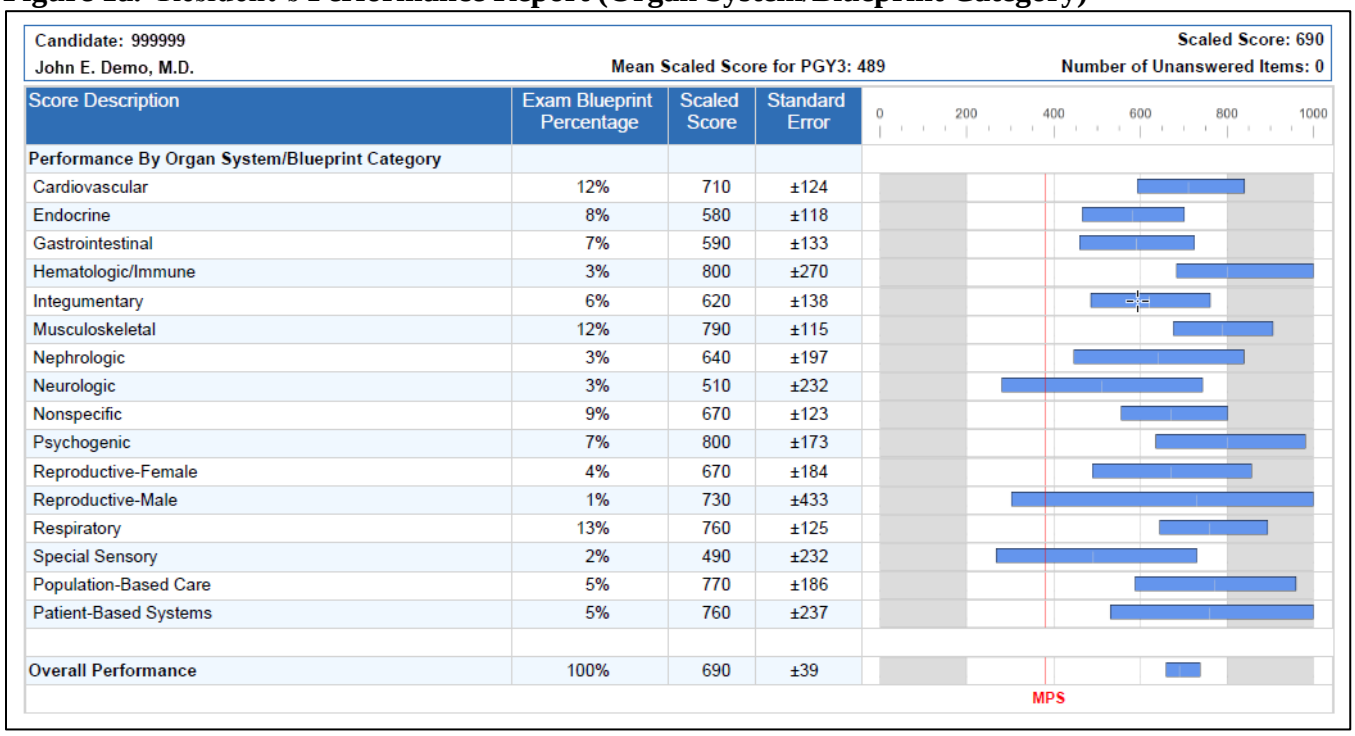
Scaled Score. The scaled score is a transformation of the number of correctly answered questions that adjusts for the difficulty of the test in such a way that the results of different versions of the examination have the same meaning. On the ITE and FMCE, scaled scores can range from 200 to 800. Scaled scores in specific content areas are based upon responses to questions on that topic only. The "Overall Performance" scaled score is based upon responses to all the scorable questions on the examination.

Standard Error. The standard error (SE) can be understood as being an indicator of the "thickness" of a scaled score. It is helpful to think of all scaled scores, both overall performance and in specific content areas, as having a thickness rather than as being a point. The SE can be used to create a confidence interval around the scaled score. Plus-or-minus 1 SE creates a 68% confidence interval around a scaled score, meaning that it is expected that the "true ability" scaled score will fall somewhere in that range 68% of the time. Plus-or-minus 2 (actually 1.96) SEs creates a 95% confidence interval. When content areas have only a few questions, the SE is larger, but even the scaled score for the entire examination has an SE appreciably greater than zero. Nevertheless, it is still sufficiently small to make a score precise enough to be useful.

Bar Chart. The bar chart on the right side of the report depicts the data from the table on the left. For each content area, the resident's performance is represented by a bar. The midpoint marker of the bar represents their scaled score. The entire bar represents the 68% confidence interval (± 1 SE) around their scaled score. When reviewing a resident's performance by content area, please consider the precision of the

scaled score. Also, be advised that the ABFM only reports scores between 200 and 800. If a resident's score is below 200 or above 800, it will be reported as 200 or 800, respectively.

Figure 1a. Resident's Performance Report (Organ System/Blueprint Category)

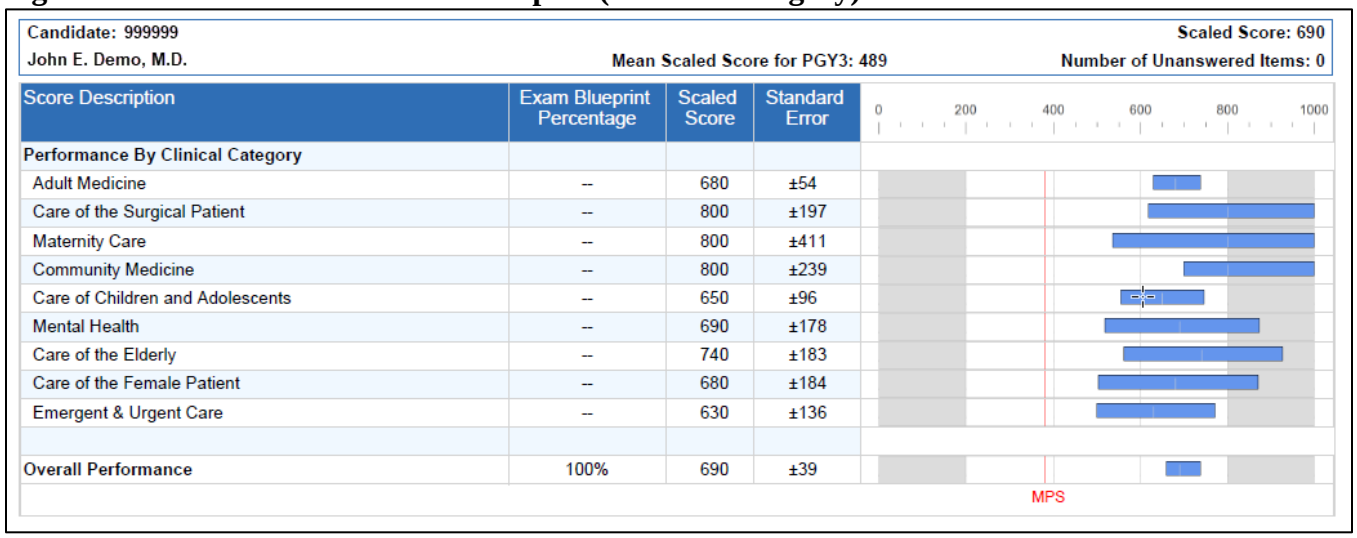


Candidate: 999999 — Scaled Score: 690
John E. Demo, M.D. — Mean Scaled Score for PGY3: 489 — Number of Unanswered Items: 0

Score Description	Exam Blueprint Percentage	Scaled Score	Standard Error
Performance By Organ System/Blueprint Category			
Cardiovascular	12%	710	±124
Endocrine	8%	580	±118
Gastrointestinal	7%	590	±133
Hematologic/Immune	3%	800	±270
Integumentary	6%	620	±138
Musculoskeletal	12%	790	±115
Nephrologic	3%	640	±197
Neurologic	3%	510	±232
Nonspecific	9%	670	±123
Psychogenic	7%	800	±173
Reproductive-Female	4%	670	±184
Reproductive-Male	1%	730	±433
Respiratory	13%	760	±125
Special Sensory	2%	490	±232
Population-Based Care	5%	770	±186
Patient-Based Systems	5%	760	±237
Overall Performance	100%	690	±39

0 200 400 600 800 1000
MPS

Figure 1b. Resident's Performance Report (Clinical Category)



Candidate: 999999 — Scaled Score: 690
John E. Demo, M.D. — Mean Scaled Score for PGY3: 489 — Number of Unanswered Items: 0

Score Description	Exam Blueprint Percentage	Scaled Score	Standard Error
Performance By Clinical Category			
Adult Medicine	--	680	±54
Care of the Surgical Patient	--	800	±197
Maternity Care	--	800	±411
Community Medicine	--	800	±239
Care of Children and Adolescents	--	650	±96
Mental Health	--	690	±178
Care of the Elderly	--	740	±183
Care of the Female Patient	--	680	±184
Emergent & Urgent Care	--	630	±136
Overall Performance	100%	690	±39

0 200 400 600 800 1000
MPS

Resident's Item Performance Report

The Resident's Item Performance Report is also presented across Organ System/Blueprint Categories (Figure 2a) and across Clinical Categories (Figure 2b). The questions are identified using their sequence number from the test. Those questions answered incorrectly by the resident are shown in bold, red font. Those questions answered correctly are shown in regular, green font. This report can be used to help residents understand the types of questions typical in each of the content categories and to identify content helpful in preparing for the certification examination. Additionally, candidates may use the questions in bold red to identify content areas in which they could gain the most from studying.

Please note that questions marked with a "P" following the question number (e.g., 100P) indicate that the question was deleted from scoring for psychometric reasons and should not be used when scoring the ITE by hand using Table 3. However, even though psychometrically deleted questions did not function as expected, reviewing them may still have some utility for an individual resident and so they have been included on these reports. When a question is deleted from the examination for content reasons, such as when there were multiple correct answers or the question was ambiguous, then it is excluded from scoring and the resident can ignore it entirely.

Figure 2a. Item Performance Report (Organ System/Blueprint Category)

	Respiratory	Cardiovascular	Musculoskeletal	Gastrointestinal	Special Sensory	Endocrine	Integumentary	Neurologic
1000		140 156		016	190			
950								
900			027					
850							111	020
800			131					
750								
700								
650		048 132	164	104 135 179		098		
600	108 180							
550	025 141	001				017	068	
500	066	172	171					078
450	188	023 061 166	122 147			182	036 162 169	
400		123 181		086 120		065		184
350	032 101		033 058	003 031	038 095	028 114		
300	002	077 091	102		136	139	186	
250	160 200	150 189	090			043 130	137	045
200	070 076 117 155 168 173		051 154	046		163	024	
150	014	013	005 015 082 116 187	063 149 165		081	079 097	
100		199	042 177	026		056 145 170 195	010 050	
50	134 148	085 107	075	193				
0	041 049 060 084 093 124	030 039 069 099 119	109 142 197			006		113 143

Figure 2b. Item Performance Report (Clinical Category)

	Adult Medicine	Care of the Surgical Patient	Maternity Care	Community Medicine
1000	016 140 190			
950				
900	027			
850				
800	052			
750	074			083
700	158			
650	048 098 104 110 132 135 164 179			
600	064 108 180			
550	017 018 025 068 141			
500	035 066 078 157 171			198
450	036 061 115 122 147 162 169 182 188	166	192	
400	065 120 123 128 161 184			133
350	003 028 032 033 058 094 095 101 114 125			
300	002 077 091 102 136 139			
250	043 045 073 090 106 130 150 189 200			
200	024 046 051 070 117 121 154 163 168 173 191		057	
150	005 014 063 079 081 082 097 126 149 165 187			
100	009 010 012 026 042 050 053 062 145 170 195 199		153	
50	007 021 075 085 089 107 134 148 193			019
0	006 008 011 030 037 039 041 044 059 060 084 099 109 113 119 124 142 143 144 174 176 178 194 197			087 100

Suggestions for Reviewing Scores with Residents

1. Primarily attention should be paid to a resident's overall scaled score. It is based on the resident's responses to all the scored questions and is the best estimate of how the resident would perform on the FMCE at that point in time. Identifying residents who need more assistance early on in residency allows for more time to get them back on track. The Bayesian Score Predictor (BSP, Section V) can also help to identify them.
2. Because a resident's scaled score is reported on the same scale across all years of residency as well as on the certification examination, using these scaled scores to monitor a resident's progress throughout training will provide an effective way for residents to get timely feedback. Resident performance should be compared to the minimum passing score (MPS) for the FMCE. Presently, the MPS is 380. Typically, ITE score can be expected to jump approximately 40-60 points from PGY1 to PGY2 and another 25-35 points from PGY2 to PGY3. Third year residents that score a 440 or better generally have a very good probability of passing the certification examination.[1-2]
3. To identify the areas of study that will produce the largest gains for them, residents should identify the content areas that have the most questions. This information is available on the score report under *Exam Blueprint Percentage* column. Of course, this should not be construed as meaning that the resident should neglect the other areas of study.
4. After ABFM has released the ITE questions to the residency programs, it is useful for the resident to go through the ITE question-by-question and
    a. remember what answer they chose and why,
    b. look up the correct answer in the critique,
    c. if the resident's answer was wrong, identify what was the *disconnect* between their rationale and the rationale in the critique,
    d. review the disconnects to see if any common themes emerge. **Note:** The disconnect could be a deficit in knowledge or it could be related to a test-taking style issue, such as not reading the question carefully, reading too much into the question, coming to a conclusion too quickly, answering a similar question, but not exactly the question asked, etc.
    e. Create a study plan based upon disconnects.
5. Do not rely heavily on subscores. ITE subscores are not sufficiently precise to reliably confirm knowledge deficiencies within a content area. This may be readily understood by examining the very large standard error associated with many of the content categories. However, these scores may be useful in the context of generating (but not confirming) hypotheses about an individual's strengths and weaknesses of knowledge.

[1] O'Neill TR, Li Z, Peabody MR, Lybarger M, Royal KD, Puffer JC. The Predictive Validity of ABFM's In-Training Examination. *Fam Med.* 2015;47(5):349-356.
[2] O'Neill TR, Peabody MR, Song H. The predictive validity of the National Board of Osteopathic Medical Examiners' COMLEX-USA examinations with regard to outcomes on American Board of Family Medicine examinations. *Academic Medicine*. 2016;91(11):1568-1575.

SECTION II

Program-level Reports

Overall Scaled Score Distribution Graphs

In Figure 3, the 2023 ITE reported scaled scores for the entire population of residents enrolled in ACGME-accredited programs are represented with a panel specific to each training year and a panel representing the combined population across all training years. The reported scaled scores increase in increments of 10. Scores lower than 200 are reported as 200 and scores greater than 800 are reported as 800. Generally, the panels show a normal, symmetrical distribution (with PGY4 as exception due to small sample size). ACGME-accredited family medicine residencies are typically a 3-year curriculum. The PGY4 chart in Figure 3 also represents the reported scaled score of those PGY4 residents in a Length of Training (LoT) pilot program. It does represent those residents who are in a dual-specialty program (Family Medicine & Psychiatry, Family Medicine & Internal Medicine, etc.)

Mean Scaled Score Graph

Each program's *Mean Scaled Score Graph* (Figure 4) shows the mean scaled score for the population and then mean scaled score for the program within each training year as well as for all training years combined.

Eight Year Trend Chart and Data

The 8-year trend summary data from Figure 5 is visually depicted in Figure 6. Each data point in the table (Figure 5) represents the height of a bar on the graph, making trends in the performance of your program both across PGY cohorts and overtime within cohorts more obvious. Comparisons to national averages are also presented. The data table presented happens to be the Microsoft Excel version, but a comma-delimited version is also available, in case you have access to or a preference for other spreadsheet software.

The graph (Figure 6) shows a program's mean scaled scores over the past eight years. Please note that programs new to the ITE may not yet have ITE scores for all years, but the national averages are still presented. Within each calendar year, your program's average score is shown adjacent to the national average for that group. Within calendar year, the mean scores for both your program and the nation are presented broken out by each training year (PGY1, PGY2, PGY3, and PGY4) and aggregated across PGY1 to PGY4. Please note a typical family medicine residency is 3 years long; however, there are several residencies that are participating in the Length of Training (LoT) pilot which provides 4 years of family medicine training. The PGY4 cohort represents only these residents. Although the graph is aesthetically busy, it provides a great deal of information.

Figure 3. Scaled Score Distribution

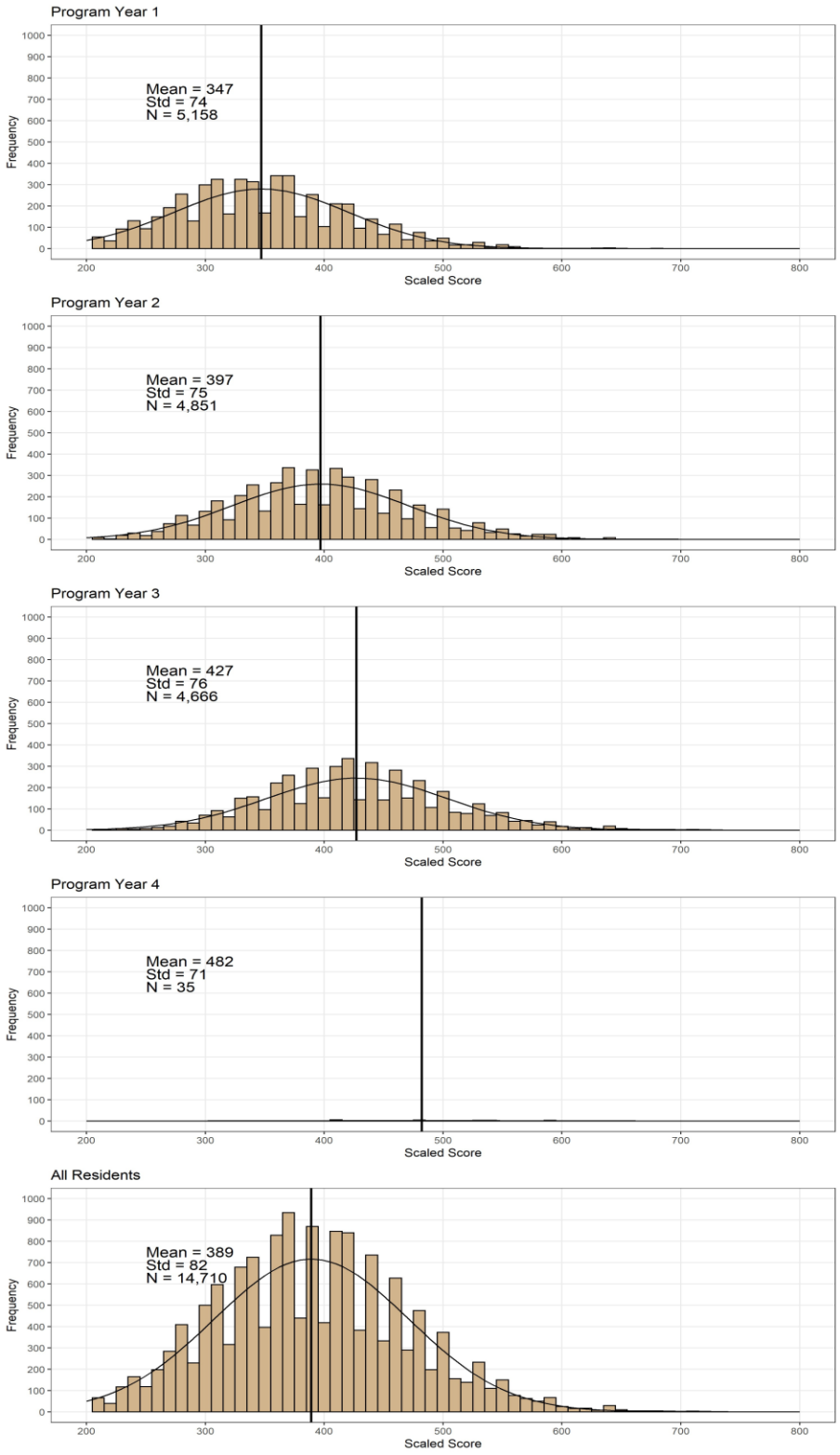


Figure 4. Example of a Program-Level Report

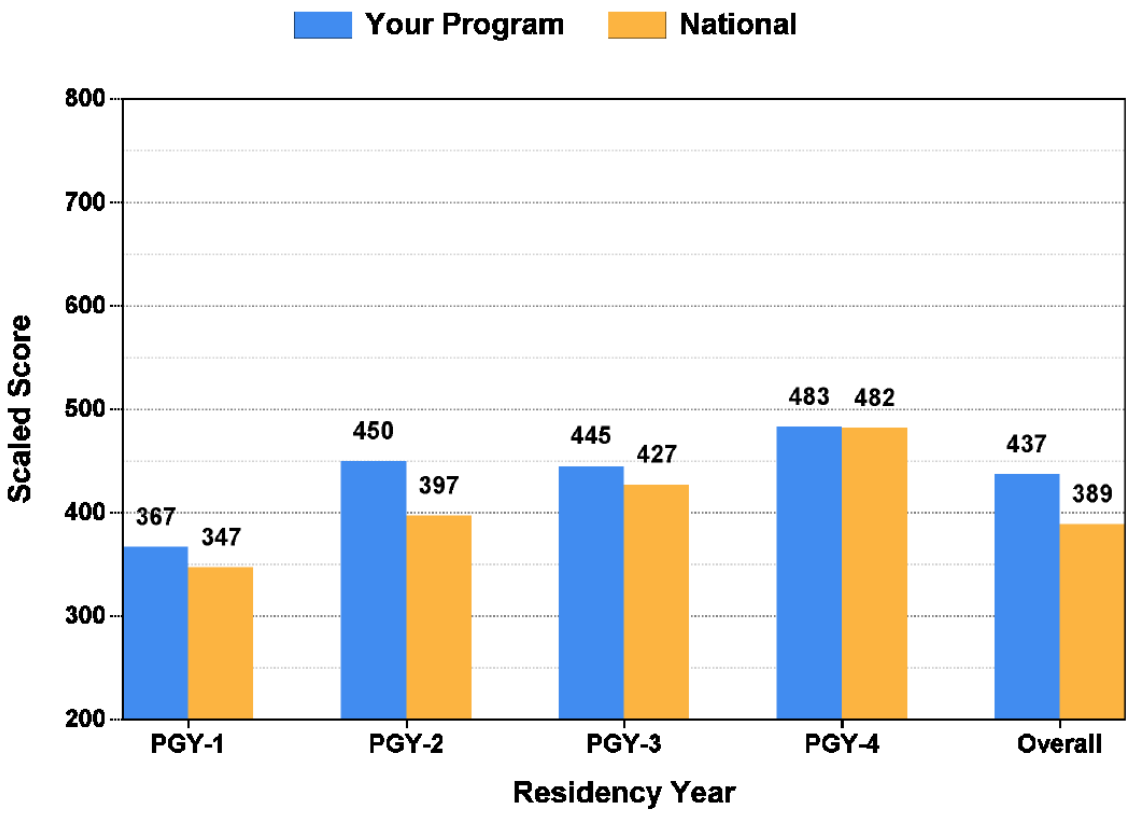
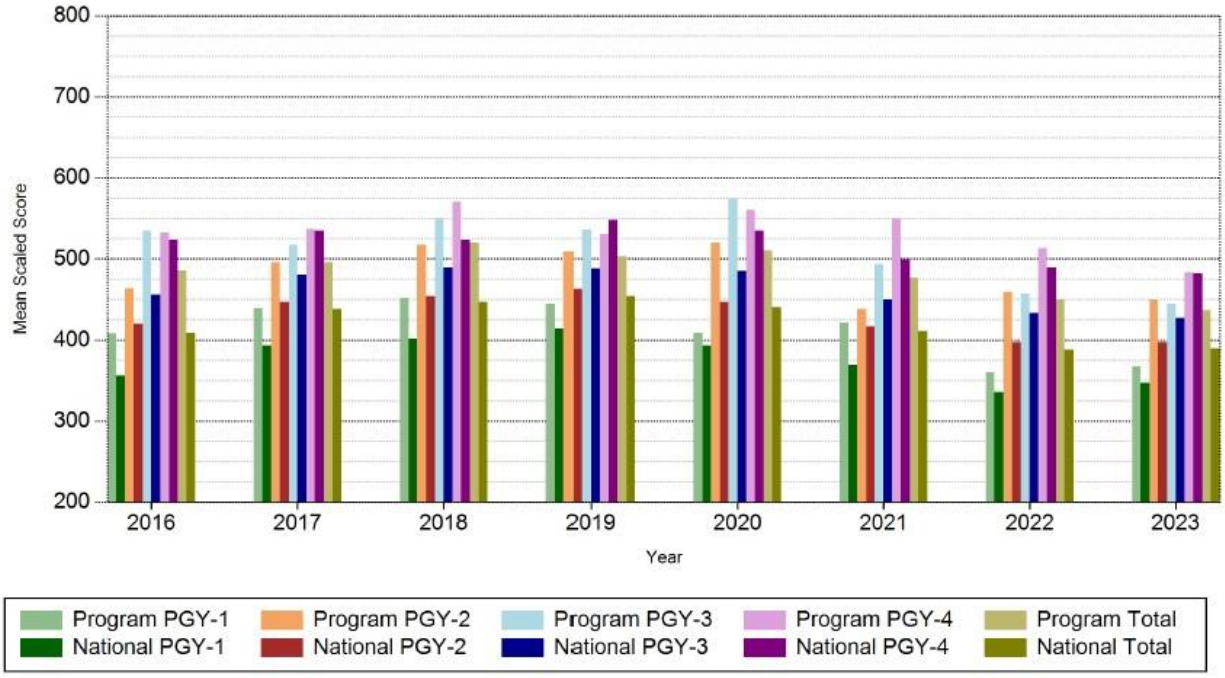


Figure 5. Data Table for 8 Year Trend Chart

	Exam Year	PGY-1	PGY-2	PGY-3	PGY-4	Total
National	2016	356	420	456	523	409
Program	2016	408	464	535	532	486
National	2017	393	447	480	535	438
Program	2017	439	496	517	537	496
National	2018	402	454	489	523	447
Program	2018	451	517	550	571	520
National	2019	414	463	488	548	454
Program	2019	445	509	536	530	503
National	2020	393	447	485	535	440
Program	2020	409	520	574	560	510
National	2021	369	417	450	499	411
Program	2021	421	438	494	550	477
National	2022	336	397	433	489	388
Program	2022	360	459	457	513	450
National	2023	347	397	427	482	389
Program	2023	367	450	445	483	437

Figure 6. ITE 8-Year Trend Chart



The ITE was scored using procedures designed to connect the ITE results to the scale of the certification examination.

SECTION III

Individual-Level Score Report Data for All Residents

Introduction

In Figures 7 and 8, examples of two data tables are presented to illustrate the values seen on each residents' detailed score report (see Figure 1). These data are provided in both Microsoft Excel and comma-delimited format. The example below uses the Excel format, which comes as two worksheets within a single workbook. The first worksheet (Figure 7) presents the data table. The second worksheet (Figure 8) shows the descriptive labels for the columns in the data table. In the comma-delimited version of this file, only the data is included. To get the labels, you must use the second excel worksheet in the workbook. The numbers seen in columns of the data table correspond to the codes in the label table. For example, columns in the data table beginning with ID101 provide values describing the "Total" set of questions on the examination, whereas columns beginning with ID201 are in regard to "Respiratory System" questions.

Row Descriptions

As seen in Figure 7, you can compare your program's performance (row 10) to the national averages for each year of residency (rows 5–9). Row 10 shows your program's ID along with data that applies to the program as a whole. Row 11 describes your first-year residents' performance followed by rows with the results for named first year residents. This pattern is repeated for the second, third, and sometimes fourth year residents in subsequent rows.

Column Descriptions

Column A shows whether a particular set of rows provides data for national values or for your program (designated by your Program ID#). Column B shows the relevant training year that is being described in each row. Column C provides residents' names. Column D provides the resident's ABFM ID number.

Columns E through I (ID101) represent performance on the entire test. Columns J through EE represent subtest performance. The same variables that are available for the entire test (ID101) is also available for the body system subtests (ID201 to ID216), as well as for the clinical categories (ID301 to ID 309). Descriptions of what these ID###s represent can be found in Figure 8.

Each consecutive set of 5 columns to the right show the number of questions in the indicated category (as in column E), the raw number of questions correct in that category (as in F), the percent correct within the category (as in G), and the scaled score (as in H) which is based upon the number correct and the difficulty of the questions in that content area, and the standard error of the scaled score (as in I).

Figure 7. Individual-Level Score Report Data Table for All Residents

2023 ITE Score Data

ITE Year: 2023

ACGMEID	Training Year	Full Name	ABFMID	ID101 Item #	ID101 Raw Score	ID101 Percent	ID101 Scaled Score	ID101 SEM	ID201 Item #
National				198	122	61.6	389	37	26
	1			198	115	58.2	347	37	26
	2			198	123	62.2	397	37	26
	3			198	128	64.7	427	38	26
	4			198	124	62.5	399	37	26
5555555555				198	122	61.6	388	37	26
	1			198	123	62.1	390	37	26
		Doogie Howser, MD	555001	198	136	68.7	480	38	26
		Nicholas Riviera, MD	555002	198	132	66.7	450	38	26
		Julius Hibbert, MD	555003	198	114	57.6	340	37	26
		Gregory House, MD	555005	198	124	62.6	400	37	26
		John McIntyre, DO	555006	198	105	53.0	280	36	26
	2			198	119	60.1	370	37	26
		B.J. Hunnicutt, MD	555008	198	124	62.6	400	37	26
		John Zoidbert, MD	555009	198	121	61.1	380	36	26
		Leonard McCoy, MD	555010	198	123	62.0	394	37	26
		John Watson, MD	555011	198	112	56.6	330	36	26
		Marcus Welby, MD	555012	198	118	59.6	360	37	26
	3			198	132	66.7	450	38	26
		Charles Emerson Winchester III, MD	555015	198	125	63.1	410	37	26
		Frank Burns, MD	555016	198	125	63.1	410	37	26
		Henry Blake, DO	555017	198	122	61.6	390	36	26
		Sherman Potter, MD	555018	198	125	63.1	410	37	26
		Benjamin F. Pierce, MD	555019	198	134	67.7	460	38	26
		Yuri Zhivago, MD	555020	198	131	66.2	440	37	26
	4			198	123	62.1	390	37	26
		Stephen Strange, MD	555015	198	114	57.6	340	37	26
		Abbie Lockhart, DO	555016	198	132	66.7	450	38	26
		Mark Green, MD	555017	198	122	61.6	390	36	26

Figure 8. Labels for the Data Table

Item Name	Description
ID101	Total Questions
ID201	Respiratory
ID202	Cardiovascular
ID203	Musculoskeletal
ID204	Gastrointestinal
ID205	Special Sensory
ID206	Endocrine
ID207	Integumentary
ID208	Neurologic
ID209	Psychogenic
ID210	Reproductive: Female
ID211	Reproductive: Male
ID212	Nephrologic
ID213	Hematologic/Immune
ID214	Nonspecific
ID215	Population-based Care
ID216	Patient-based Systems
ID301	Adult Medicine
ID302	Care of the Surgical Patient
ID303	Maternity Care
ID304	Community Medicine
ID305	Care of Children and Adolescents
ID306	Mental Health
ID307	Care of the Elderly
ID308	Care of the Female Patient
ID309	Emergent & Urgent Care

SECTION IV

INSTRUCTIONS FOR SCORING BY HAND

Introduction

Scoring the ITE by hand requires that you have answered ALL the questions on the examination. Because there is noticeable variability in the difficulty of the questions, one must consider BOTH how many questions were answered correctly and how difficult the questions were. The raw to scaled score conversion table (Table 3) applies only to ENTIRE test. There are no conversion tables available to the end user that are specific to subsets of questions selected by the end user.

Key Changes and Questions Removed from Scoring

A psychometric evaluation of each question is performed following the initial administration of each exam form. Whenever a question is not functioning as expected, the question, critique, references, and answers are carefully reviewed by a content expert prior to the final scoring of the examination. For the 2023 ITE, no answers had to be re-keyed, no questions were deleted for psychometric reasons, and two questions were deleted for content reasons. After removing these questions, the final number of questions scored was 198.

Questions Deleted	Reason for Removal
Question 67	Content Delete
Question 129	Content Delete

Scoring by Hand

To score the examination by hand is a rather slow process; however, it is very easy and quite intuitive.

Step 1. Take the test and record the answers.

Step 2. Using the Critique Book as the answer key, mark all the questions as RIGHT or WRONG. Unanswered questions are wrong.

Step 3. Count how many questions were answered correctly, excluding any question removed from scoring (content or psychometric deletes).

Step 4. Look up in Table 3 (Raw Score to Scaled Score Conversion table), the number of questions answered correctly in the raw score column. The corresponding number in the Scaled Score column is your scaled score.

Table 3. Raw Score to Scaled Score Conversion

Raw Score	Scaled Score	Raw Score	Scaled Score	Raw Score	Scaled Score
0-91	200	125-126	410	155	620
92-93	210	127-128	420	156	630
94	220	129	430	157-158	640
95-96	230	130-131	440	159	650
97-98	240	132	450	160	660
99	250	133-134	460	161	670
100-101	260	135	470	162	680
102-103	270	136-137	480	163	690
104-105	280	138	490	164	700
106	290	139-140	500	165	710
107-108	300	141	510	166	720
109-110	310	142	520	167	730
111	320	143-144	530	168	740
112-113	330	145	540	169	750
114-115	340	146	550	170	760
116	350	147-148	560	171	770
117-118	360	149	570	172	790
119-120	370	150	580	173-198	800
121	380	151-152	590		
122-123	390	153	600		
124	400	154	610		

Table 4 is included so that people attempting to administer this exam via pencil and paper can label the questions by content category for review purposes.

Table 4. Questions by Blueprint Category

Cardiovascular	1, 13, 23, 30, 39, 48, 61, 69, 77, 85, 91, 99, 107, 119, 123, 132, 140, 150, 156, 166, 172, 181, 189, 199
Endocrine	6, 17, 28, 43, 56, 65, 81, 98, 114, 130, 139, 145, 163, 170, 182, 195
Gastrointestinal	3, 16, 26, 31, 46, 63, 86, 104, 120, 135, 149, 165, 179, 193
Hematologic/Immune	21, 53, 71, 106, 157, 191
Integumentary	10, 24, 36, 50, 68, 79, 97, 111, 137, 162, 169, 186
Musculoskeletal	5, 15, 27, 33, 42, 51, 58, 67, 75, 82, 90, 102, 109, 116, 122, 131, 142, 147, 154, 164, 171, 177, 187, 197
Nephrologic	11, 37, 62, 96, 126, 185
Neurologic	20, 45, 78, 113, 143, 184
Nonspecific	7, 18, 29, 34, 47, 55, 74, 80, 89, 103, 110, 121, 128, 138, 151, 161, 176, 194
Patient-Based Systems	9, 35, 59, 83, 94, 100, 125, 144, 178, 198
Population-Based Care	19, 44, 52, 64, 73, 87, 115, 133, 153, 167
Psychogenic	4, 22, 40, 57, 72, 92, 105, 118, 129, 146, 159, 174, 183, 196
Reproductive: Female	12, 54, 88, 112, 127, 152, 175, 192
Reproductive: Male	8, 158
Respiratory	2, 14, 25, 32, 41, 49, 60, 66, 70, 76, 84, 93, 101, 108, 117, 124, 134, 141, 148, 155, 160, 168, 173, 180, 188, 200
Special Sensory	38, 95, 136, 190
This list is specific to the 2023 ITE. It cannot be used with tests from other years.	

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SECTION V

Bayesian Score Predictor

Overview

The ABFM created the Bayesian Score Predictor (BSP) to provide program directors and residents with information to help interpret what a resident's ITE score implies about their future performance on the FMCE. The ITE score can be interpreted directly as our best estimate of how the resident would perform on the FMCE at that point in time. Of course, the score is only an imprecise estimate. Given that the standard error of measurement (SEM) for the ITE is roughly 38 points, the observed ITE score has a 68% chance of being within 38 points of the resident's true ability and a 95% of being within 76 points of the resident's true ability. The imprecision described by the SEM is just an articulation of the bounds of the random sampling process. It does not consider any additional factors (fatigue, illness, extraordinary distractions, etc.) that might make the observed ITE score a less accurate estimate of the resident's true ability. With regard to predicting whether an individual will pass the FMCE, ITE scores over 440 are very re-assuring, assuming that the ITE was taken under test-like conditions. By test-like conditions, we mean that (1) the examinee has not seen the ITE questions prior to taking the test, (2) the examinee takes the test within the allotted time constraints, and (3) the examinee does not use external resources, including people, to look up the answer to the questions. For additional information regarding the predictive value of the ITE and the COMLEX-USA for predicting performance on the FMCE, please see the citations in the footnotes[3,4].

Although directly interpreting the ITE score has the advantage of providing an estimate of how the examinee would perform on the FMCE, it does not provide a probability of passing. For example, if 380 is passing for the FMCE and a PGY3 resident scored 380 on the ITE, what is the probability of them passing the FMCE? It is not 100%! It is more like 85%[5]. Also, consider the case of a second-year resident who gets an ITE score of 350. Is this resident on track to pass the FMCE? The ABFM Bayesian Score Predictor (https://rtm.theabfm.org/bayesian/predictor) can help answer these questions.

Using the Bayesian Score Predictor

The Bayesian Score Predictor (BSP) is intended to provide family medicine residents and program directors with an estimate of the resident's probability of passing the ABFM cetification examination when they take it at the end of PGY-3. The program's user interface (Figure 9a) has six boxes. The first box contains either a range of COMLEX-USA® Level 2 CE scores or a range of USMLE Step 2 CK scores with the expected distribution of examinees for those scores. You can choose which one you wish to enter. The next three boxes contain the useful range of scores

[3] O'Neill TR, Li Z, Peabody MR, Lybarger M, Royal KD, Puffer JC. The Predictive Validity of ABFM's In-Training Examination. *Fam Med*. 2015;47(5):349-356.

[4] O'Neill TR, Peabody MR, Song, H. The Predictive Validity of the National Board of Osteopathic Medical Examiners' COMLEX-USA Examinations With Regard to Outcomes on American Board of Family Medicine Examinations. *Acad Med*. 2016; 91(1): 568–1575.

[5] Please note that in Section I, we indicated that someone who just barely earned a score of 380 would theoretically have a 50% chance of passing if they took the examination at that point in time. Here we are discussing, everyone who earned a 380 (just barely to almost a 390) and assuming that they still have an additional 5 months in their residency program before taking the certification examination.

on the ITE, one box for each year of residency. The next box, the fifth one, represents the useful range of scores on the FMCE and the distribution of examinees associated with these scores. The sixth and final box indicates the probably of passing the FMCE. To really understand what the BSP is doing, it helps to play with it for a while. Do not worry about breaking it. It is a web-based application, so it is difficult to break.

In its initial state, the boxes are populated with a useful distribution of licensing examination, ITE, and FMCE scores (Figure 9a). However, if the user clicks on a score in any one of these boxes (Figure 9b) the distribution changes from a distribution of scores to the selected observed score. This will in turn cascade forward to revise the distributions to reflect the expected ITE scores in later years of residency and then to the expected distribution of FMCE scores. These revisions represent the probabilities that are expected based upon only those people who scored the selected score. Please note that when you click on a score to indicate that it is the score you earned; the observed score has a 100% probability of having happened because you said that it did. This directs the BSP to consider only the distributions that would result from the population of people who earned that observed score.

As an example, consider the case of a resident who scored 270 in PGY-2 (Figure 9b). Here, the user would click on <270> in the PGY2 box. This will (1) change the PGY2 distribution to 100% for the score of 270 and (2) change the expected score distributions for PGY3 and for the FMCE to reflect the percentages expected for a resident who scored a 270 in PGY2. You can see that this also updates the resident's probability of passing.

Furthermore, if the user later clicks on a different PGY2 score, say 410 (Figure 9c), the program will change the "observed score" to 410 and update the distributions and probabilities accordingly. A second click on 410 restores the original default distribution for PGY2. It is important to note that if a score is selected for PGY1 and then for PGY2, the PGY2 score overrides the PGY1 score because it occurred at a point in time closer to when the FMCE pass-fail decision is made.

As another example, let us say that a resident in PGY1 earned a 340 on the ITE. Clicking on <340> in the PGY1 Score column will show the expected future score distributions. This will also cause the other four boxes re-populate and the resident's expected pathway can be understood as the mean score found in PGY2, PGY3, and on the FMCE. The acceptability of that expected pathway is really a question of how satisfied the resident and program director are with the probability of passing the FMCE.

It is also important to note that probabilities cascade backward as well as forward, showing the distribution of scores that likely generated the observed score. This is not helpful for making predictions about passing, but it might be helpful in quantifying the degree of unexpectedness when something strange happens, such as a low scorer in PGY 1 and 2 scoring unexpectedly high in PGY3.

If you have questions regarding the Bayesian Score Predictor, please contact the ABFM psychometrics department.

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Figure 9a. Bayesian Score Predictor in its Initial State

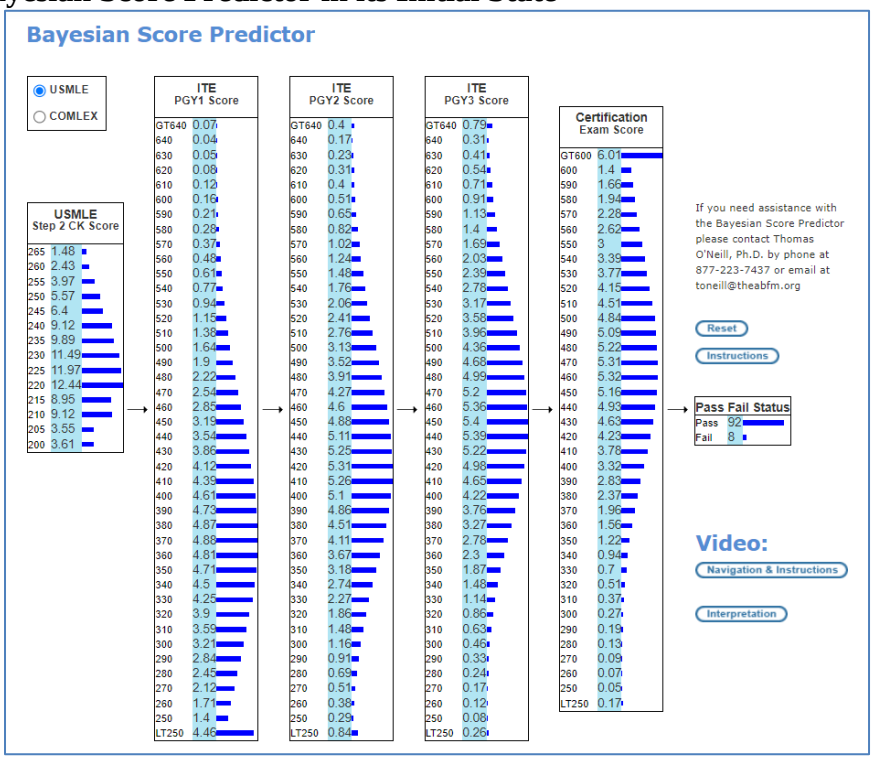


Figure 9b. Bayesian Score Predictor with PGY2 = 270

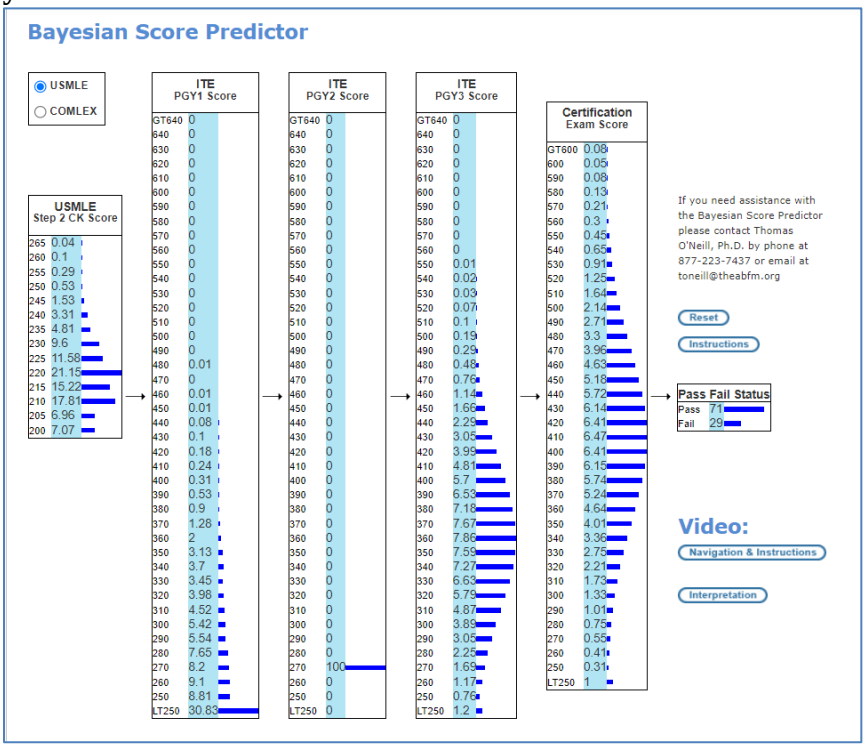
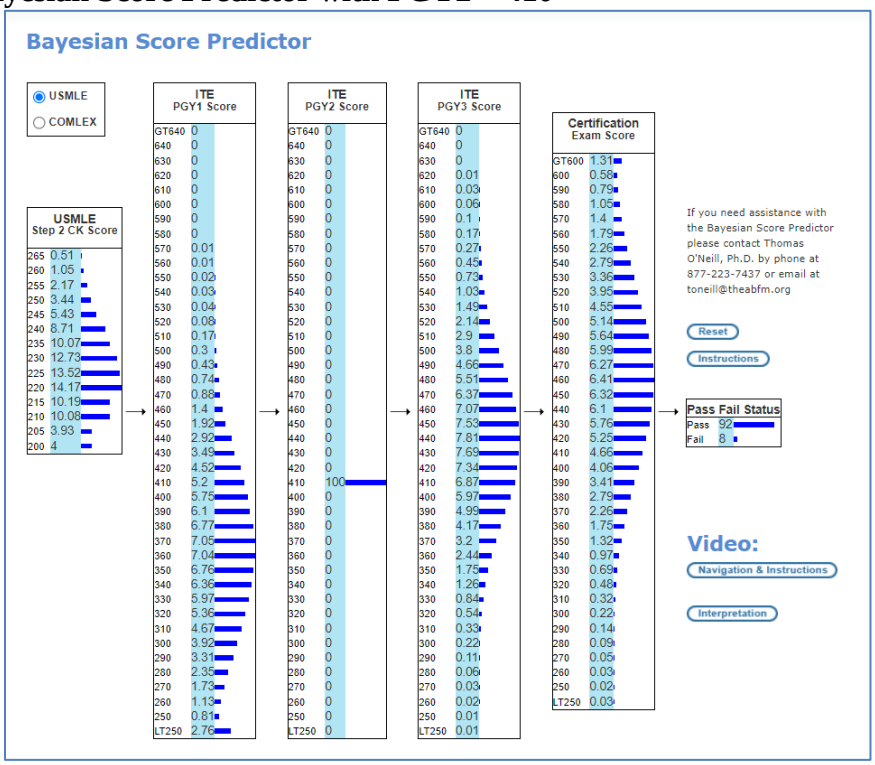


Figure 9c. Bayesian Score Predictor with PGY2 = 410



SECTION VI

Psychometric Summary

Purpose

The ABFM offers residents enrolled in ACGME-accredited residency programs the opportunity to take the ITE. The purposes of the ABFM's ITE are: (1) to provide each resident with a low-cost, low-stakes opportunity to become familiar with the general format and item writing style that will be used on the FMCE and (2) to provide each resident and his or her program director with an opportunity to assess how well the resident is progressing toward eventually passing the FMCE. As the ITE is designed as a low-stakes examination, the ABFM advises program directors that the results should not be used to make important decisions related to the promotion or advancement of residents taking the exam. To achieve the purposes stated above, it is important that: (1) the ITE content be similar to that of the FMCE, (2) the ITE results be a good approximation of how a resident would have performed on the FMCE at that point in time, and (3) the ITE be predictive of an examinee's future results on the FMCE.

It is important to note that ABFM has conducted research to connect ITE scores with the ABFM's certification exam results so that reasonable predictions can be made. However, the research is silent about making predictions about how well the ITE predicts performance on the American Osteopathic Board of Family Physicians (AOBFP) certification exam. Similarly, our research is also silent on how well the American College of Osteopathic Family Physician's (ACOFP) In-Service Exam (ISE) predicts performance on ABFM certification examination. ABFM just does not have the data for those analyses.

Criterion-related Validity

The ABFM's ITE was designed to have a high degree of concurrent and predictive validity, both of which are forms of criterion-related validity. The criterion in both is performance on the FMCE. The concurrent validity claim for the ABFM's ITE is that it is intended to produce scores that would be predictions of how an examinee would perform on the FMCE if he or she had taken it instead of the ITE at that point in time. To achieve this, each form of the ITE is built to the same specifications as the FMCE and the ITE scores are equated onto the FMCE scale using the dichotomous Rasch model and a common-item equating design. Concurrent validity with the FMCE is partially established through the regular quality checks that ensure the ITE is developed with the correct content specifications and that the equating was successful. Because the results of these quality control checks have always been quite good, the ABFM has never conducted an experiment in which both tests were administered to examinees on consecutive days; however, the ABFM does administer the FMCE a few months after PGY3 residents take the ITE, which is how the ABFM usually establishes the predictive validity of the ITE[6].

Test Specifications

The ITE contains 200 multiple-choice questions and is built to mirror the same content domain proportions as the FMCE. The process used to develop the content specifications for the FMCE is

[6] O'Neill TR, Li Z, Peabody MR, Lybarger M, Royal KD, Puffer JC. The Predictive Validity of ABFM's In-Training Examination. *Fam Med*. 2015;47(5):349-356.

described in greater detail by Norris, Rovinelli, Puffer, Rinaldo, and Price[7]. Since 2013, it has been administered only in a computer-administered format. Each year, there is a different form of the ITE with no questions in common from previous forms because after each administration, the questions on that test are made available to residents and diplomates for educational purposes. These questions are then no longer secure and would probably be known by the examinees if they were reused.

Equating the ITE to the FMCE

To equate the ITE across administrations and to make the ITE score represent the examinee's predicted performance on the FMCE, the ABFM includes a small number (approximately 40) of ITE questions as unscored pretest questions on the FMCE. These questions are then calibrated onto the FMCE scale. Displacement[8,9] indices are used to check that the relative question difficulty calibrations derived from the FMCE examinee population still hold for the ITE examinee population. In cases where the displacement is more than 0.6 logits, the question is no longer considered an anchor and the ITE data is used to recalibrate it. In this way, these ITE questions and their associated FMCE calibrations are used to connect each administration of the ITE to the continuously maintained Family Medicine Certification - Scale (FMC-Scale).

Scoring

Prior to scoring, all 200 questions are subjected to an item analysis to flag potentially problematic test questions. Questions that are determined to be inadequate for content reasons are removed from scoring. Questions that are determined to psychometrically function in a manner that degrades the measurement quality of the ITE are also excluded. These questions are listed in Section IV.

The remaining ITE questions are scored as right or wrong and these scored responses are transformed into a logit metric using the dichotomous Rasch[10] model. Next, the logit values are converted into scaled scores that range from 200-800. Scores below 200 are set to 200 and scores above 800 are set to 800. The Rasch model, in conjunction with a common item equating design, is also used to equate examinations across test forms and years of administration. The use of a common scale with a passing standard that is held constant for useful periods of time has the advantage of providing a more stable target for making predictions related to whether a candidate will pass or fail.

2023 Administration

The ABFM's ITE is administered to nearly all family medicine residents in ACGME accredited programs. In 2023, there were 14,710 residents from 719 residency programs that took the ITE. The number of residents in each year of residency was evenly distributed.

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[7] Norris TE, Rovinelli RJ, Puffer JC, Rinaldo J, Price DW. From Specialty-Based to Practice-Based: A New Blueprint for the American Board of Family Medicine Cognitive Examination. *Jounal of the American Board of Family Practice*. 2005;18(6):546-554.

[8] Linacre, J. M. (2013). *Winsteps® Rasch measurement computer program User's Guide*. Beaverton, Oregon: Winsteps.com.

[9] O'Neill, TR, Peabody, MR, Tan RJB, Du, Y. How much item drift is too much? *Rasch Measurement Transactions*, 2013, 27:3 p. 1423-4.

[10] Rasch G. *Probabilistic models for some intelligence and attainment tests*. Copenhagen, Denmark.: Danish Institute for Educational Research; 1960.

Reliability and Person/Item Fit

When a test is administered, it is desirable to have some assurance that comparable results would be obtained if the same people were tested again under similar conditions. This reproducibility of test scores is referred to as reliability. It is important to note that reliability is not a property of the test, but rather a property of the data set. It is influenced by both the sample of candidates and the test. When using the Rasch model, it can be thought of as how well the questions spread out the people (person reliability, the conventional usage) and how well the people spread out the questions (item reliability, sometimes considered by content developers, but not by examination end-users). Given the data from the 2023 ITE, 1.95 strata of candidates (Table 5) were discernible, which is comparable to a reliability of 0.79. With regard to the questions, 55.85 strata (Table 6) were discernible, which is comparable to a reliability of 1.00.

Model fit refers to how well the data conform to the requirements of measurement. This is considered both for examinees and for questions. A summary of fit can be found in Table 5 for candidates and Table 6 for questions. Four statistics are considered: Information Weighted Mean Square (INFIT MNSQ); Information Weighted Mean Square Standardized (INFIT ZSTD); Unweighted Mean Square (OUTFIT MNSQ); and Unweighted Mean Square Standardized (OUTFIT ZSTD).

The mean square statistics have an expected value of 1.0, but can range from 0 to infinity. They can be thought of as an indicator of the magnitude of the disturbances in the data. The standardized mean square statistics have a mean of 0 and a standard deviation of 1. They can be thought of as indicators of whether the disturbance could be attributed to chance. The weighted statistics are more sensitive to disturbances near the person's ability (or the question's difficulty). The unweighted statistics are more sensitive to outlying disturbances such as high ability people missing an easy question or a low ability people making more lucky guesses than expected.

The magnitude of disturbances among the questions did not rise to a level to warrant concern; thus, the issue regarding whether the disturbance is attributable to chance is irrelevant. There were some questions that exhibited larger than preferred levels of misfit; however, those questions also had high levels of displacement, but did not reach the threshold at which they would be unanchored (0.6 logits).

Table 5. Summary of 2023 ITE Candidates (N = 14, 710)

	Ability[1]	SE	INFIT MNSQ	INFIT ZSTD	OUTFIT MNSQ	OUTFIT ZSTD
Mean	394	37	1.01	0.11	1.00	0.05
S.D.	86	1	0.08	1.05	0.17	1.03
Max	735	48	1.43	4.92	2.62	6.39
Min	-120	36	0.75	-3.64	0.59	-2.67

Note: Separation, the ratio of the precision of the candidate ability estimates to the adjusted spread of ability among candidates, was 1.95. This is a ratio measure of separation in Root Mean Square Error units. The separation expressed as a reliability coefficient (comparable to a KR20 or Cronbach's alpha) was 0.79.

[1]This is based on the true ability, no ceiling/flooring limits

Table 6. Summary of 2023 ITE Questions (N = 198)

	Difficulty	SE	INFIT MNSQ	INFIT ZSTD	OUTFIT MNSQ	OUTFIT ZSTD
Mean	256	5	1.01	0.69	1.00	0.53
S.D.	311	2	0.08	4.83	0.08	5.20
Max	1151	22	1.60	9.90	1.36	9.90
Min	-714	4	0.77	-9.90	0.74	-9.90

[*]Separation, the ratio of the precision of the question difficulty estimates to the adjusted spread of difficulty among questions, was 55.85. This is a ratio measure of separation in Root Mean Square Error units. The separation expressed as a reliability coefficient (comparable to a KR20 or Cronbach's alpha) was 1.00.

END OF REPORT